

Sleeping Machines: what is known

Computing in time with races, holds and vetoes · report, 28 September 2026

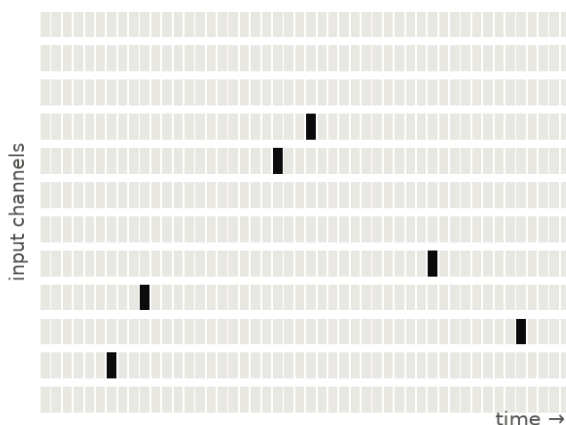
In plain terms

Today's neural networks are **clocked and dense**: at every step, every input is multiplied by every weight, whether or not anything happened. Many real signals are the opposite: long silences broken by precisely timed events (nerve spikes, trades on a market, sensor alarms), where *when* something happens is the information.

Sleeping Machines are networks that only work when an event arrives. A node waits. It fires when the right inputs arrive in the right time window ("B within 1.5 s after A"), and the first node to fire gives the answer, a *race*. Silence costs nothing, and time itself does the computing: a delay or a waiting window plays the role that a weight matrix plays in a dense network. A signal can also carry a small vector (a few numbers), and its content sets its own delay: when it arrives decides how much it counts.

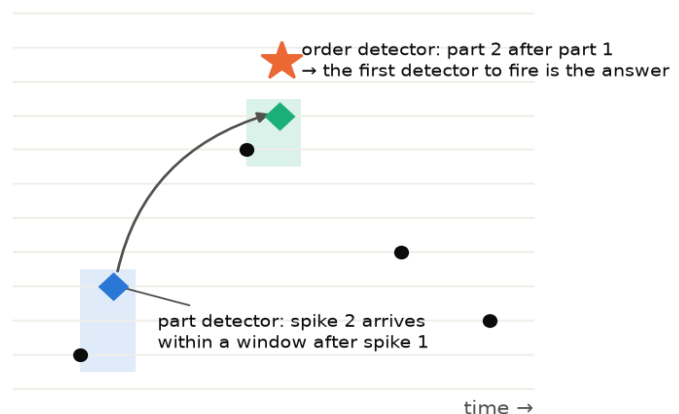
The questions are whether such networks can **learn** (with credit that flows only along the events that actually happened: a node adjusts only its few connections that were active, like moving money between accounts under a fixed budget) and whether they can **match or beat** MLPs and Transformers.

Clocked network: every cell, every tick



576 cells × every weight $\approx 10^5$ – 10^6 multiply-adds

Event network: work only when a spike arrives



≈ 10 events; silence costs nothing

Highlights

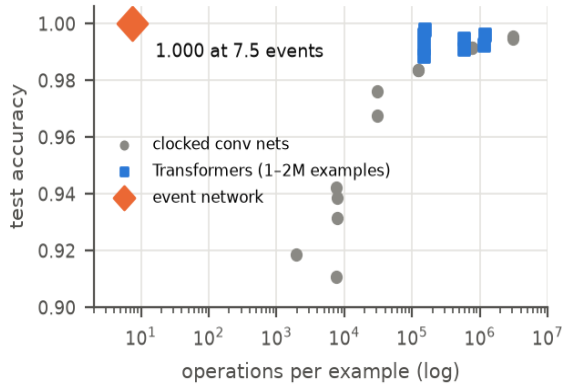
- **Same accuracy, 10,000–100,000× less computation.** On timing-pattern recognition a learned event network is perfect (1.000) using ≈ 7.5 events per example; Transformers reach 0.989–0.998 at 150k–1.2M multiply-adds after 1–2M training examples.
- **It groks where a Transformer does not.** Trained on 30% of all (a, b, c) triples, it learns (a + b + c) mod 17 and is 99.4–99.9% correct on the triples it never saw; a Transformer with weight decay stays at 3–63%.
- **Ten times less data.** On the depth-3 order task the event network is 99.8–99.9% correct after 1,000–2,000 examples seen once; a Transformer allowed as many passes as it likes needs about 10,000–20,000 for 99% (2,000: 33–41%; 5,000: 82–90%).
- **Learning to retrieve from $\geq 250\times$ less data.** In a recall task where the network must learn which stored key a query refers to (what attention learns), race attention trained by local credit is 100% correct after 1,000–4,000 examples (5 of 5 runs) and stays 100% on contexts four times longer. Of seven Transformer configurations (width 64–128, 2–4 layers, absolute or relative positions), only the two largest solve it, after 400k–1M examples, and they reach at most 72% on the longer contexts (E61; the best configuration twice, solving it between 400k and 1M examples).
- **Deep order from a few thousand examples.** Recognizing which of 20 orders of four patterns occurred needs four levels of "this, then that". The network finds the right detectors among 55 million candidates and is 99.9–100% correct → on 5 of 5 runs after 10–15k examples, with $\approx 2,000$

learning updates, ≈ 150 events per example and only $\approx 80k$ connections ever created. **At depth 4 a Transformer makes about ten times as many errors with the same 40k examples, and still 4-8 times as many with 2M** (0.990; 0.992–0.996 vs 0.999–1.000), at $\approx 7,000\times$ the computation; at depth 3 it matches only with 8–400 $\times$ more data.

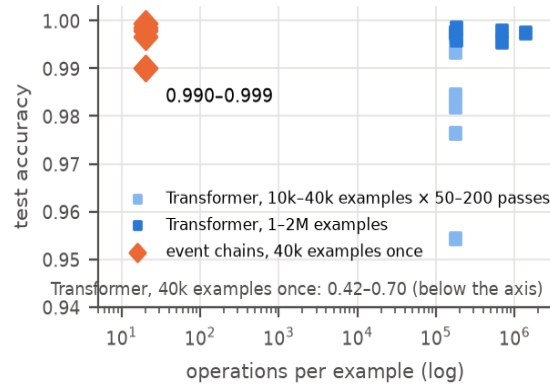
- **Composing parts: Transformer-level accuracy from one pass over the data, at $\approx 10^4\times$ less computation.** 15 classes built from ordered pairs of shared motifs: 0.990–0.999 (mean 0.9965) from 40k examples seen once, with learned timing windows, at ≈ 20 events per example; a Transformer needs 2M examples for 0.9955–0.998 and reaches 0.9935–0.9965 given the same 40k examples 50 times.
- **A world model of a real market stream within 0.08–0.19 nats of a Transformer at $\approx 1/3000$ of the computation.** On days it never saw, a small event network with slow regime counters beats a GRU point process and comes within 0.08–0.19 nats per event of a Transformer point process, at ≈ 40 operations per event instead of $\approx 110k$ –130k; the Transformer is the more accurate model.
- **Learning cost follows activity, not size.** Eight times more inputs (12 $\rightarrow$ 96 channels) costs no more learning mistakes.
- **New theory, proved:** exactly what one event node can compute and where depth is needed; why a fixed weight budget lets a node learn an AND without knowing which half was wrong; that deep order is trainable with the fewest mistakes any learner can guarantee; and that a race of clocks carries both the softmax (which fires) and its normalizer (when), so Transformer attention and its gradients are computed on average exactly by local rules: Transformers, including their training, are a limit of these networks. And event networks are exactly controlled differential equations driven by their events: the order detectors they learn are the universal features of event streams, the state-space units behind today's best event-stream models are a special case, and a race run in continuous time is more expressive than a softmax, the more so the longer it deliberates (§104). When signals carry vectors whose content sets their delays, a receiver whose state fades computes exactly softmax attention, paying only for the messages that match (§105), and the work attention then costs is set by how sharp it is, not by how much context there is (§106).

Scope of the evidence. The supremacy results (timing, composition, deep order, grokking) are on synthetic tasks built to test one capability at a time, where the target is exactly expressible by the primitives. The theory behind them (what a node computes, mistake bounds logarithmic in the candidate basis, cost proportional to events) is not task-specific: the reason to expect them to carry over to sparse, precisely timed real streams. On the real data tested so far the event network is competitive at a small fraction of the computation, not ahead: spoken digits 0.675 vs ≈ 0.70 (LSTM) and 95–96% (event-by-event state-space models); a market world model 0.08–0.19 nats behind a Transformer point process; no trading edge after fees in four markets; on a real event-camera benchmark (DVS128 Gesture) far behind: 0.70 vs 94–98% published.

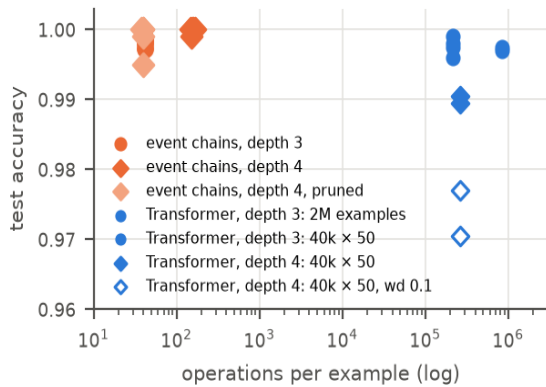
**Timing patterns:
same accuracy, $\sim 10^5 \times$ less work**



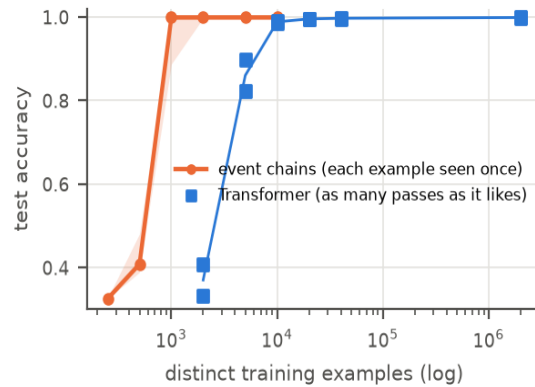
**Composing parts: Transformer-level accuracy
from one pass, $\sim 10^4 \times$ less work**



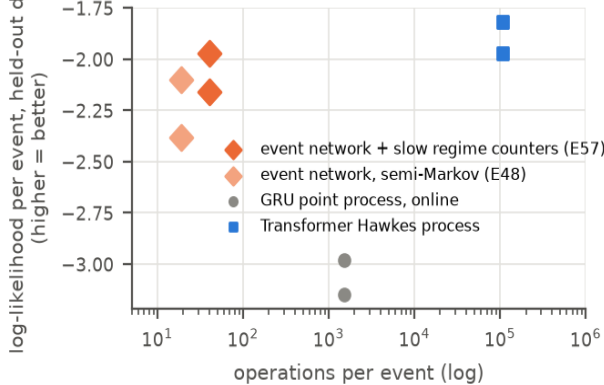
**Deep order: parity at depth 3,
 $\sim 10 \times$ fewer errors at depth 4 (equal data)**



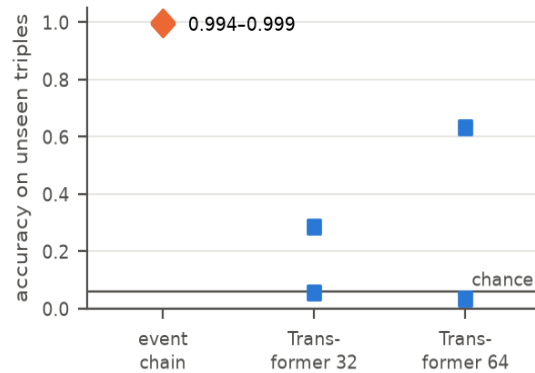
**Data efficiency (depth-3 order): 0.99 from
 ~ 1 -2k examples vs ~ 10 -20k for a Transformer**



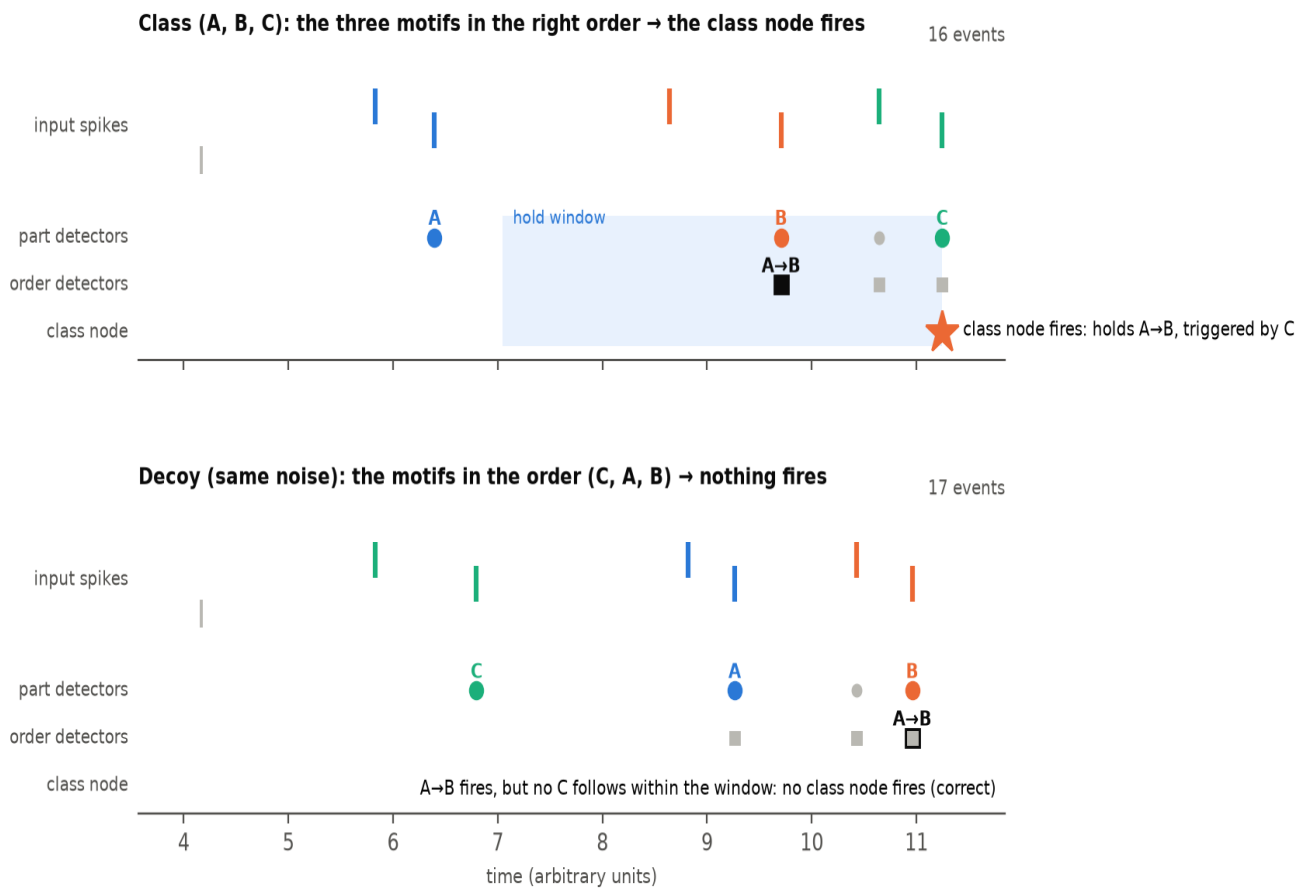
**Market world model: within 0.08-0.19 nats
of a Transformer, $\sim 1/3000$ of the work**



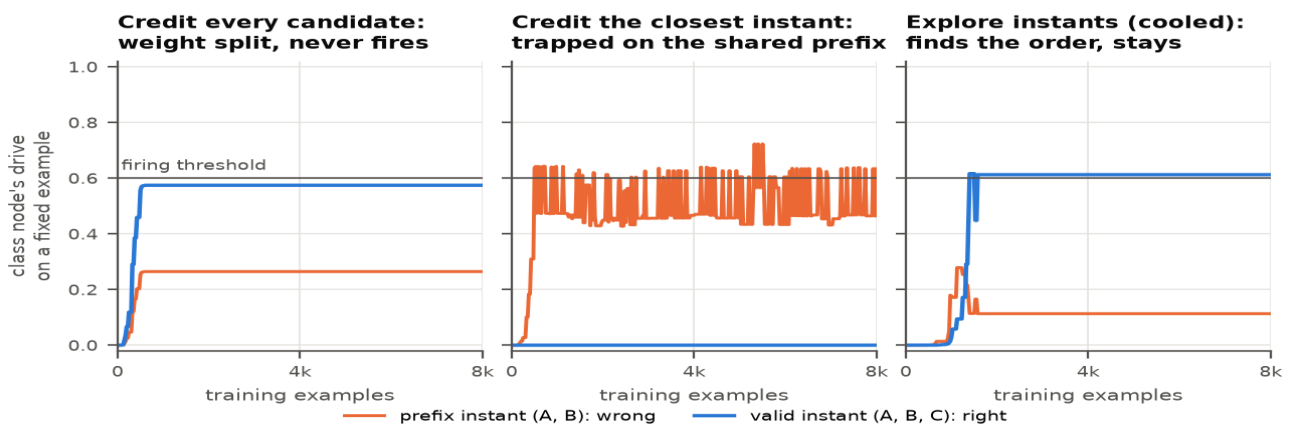
**Grokking (a + b + c) mod 17 from 30%:
the event chain generalizes**



What the network actually does on one example: spikes arrive; part detectors fire when two spikes are close enough in time; an order detector fires when part B follows part A; the class node holds that and fires when C arrives. With the same motifs in another order, the “A then B” detector still fires but nothing completes the pattern.



Why learning it is not trivial. When a detector for “A, then B, then C” fails to fire, which of its connections should change? Crediting every candidate spreads the weight so thinly that the node never fires; crediting the tempting shortcut (“A, then B” is shared with another class) traps it; exploring a little, then settling, finds the right order and keeps it (§84).



Where it does not win yet: event-camera gestures (0.70 vs 94–98% published), spoken digits (0.675 vs 0.70 for a published LSTM and 95–96% for event-by-event state-space models, whose unit the new theory identifies as a special case of ours; E74 tests a network that computes with delays and vectors together), and trading, where no learner beats buy-and-hold on this data (an audit shows why: the predictable edge, ≈ 1 bp per trade, is below any taker fee). **Next:** a path to generative language models built this way (section 10).

Sleeping Machines proposes that computation can happen **in time rather than memory**: candidate events race, the first to fire cancels the rest, and what a node computes is set by delays, by how long it holds an input, and by inhibition that arrives in time. This report states what is now known about such networks, why, and what remains open. Derivations and proofs are in experiments/THEORY.md (cited as §n).

Summary

- **What one node computes is exactly characterized, and it says where depth is needed.** A node with a trigger, hold and veto inputs accepts a product set in lag coordinates; one node orders at most three events; two layers compute every conjunction of bounded time differences, three layers every union (§71, proved; exhaustive search agrees). Nodes with state (arm/disarm) are strictly stronger.
- **Event networks learn timing with very few mistakes** if winning is positional (pull the event that should have won, never push a loser later) and order uses held intervals, not aligned delays: $O(\log N + \text{a few})$ mistakes per node.
- **On a timing task a learned event network matches or beats dense models at 10^4 - $10^5\times$ lower cost**, with nothing given: 1.000 at 7.5 synaptic events per episode vs 0.989-0.996 for event-token Transformers at 146k-1.16M multiply-adds and 0.995 for a clocked conv net at 3.07M.
- **Depth is learned natively when credit is right (§83-§86).** Summed potentials with conserved multiplicative credit, credit to one instant with cooled exploration, and synapses grown only when credited (provably the same decisions as dense weights) learn order among three and four parts: 0.999-1.000 on 5/5 seeds at depth 4 with $\approx 2,000$ updates and $\approx 80k$ grown synapses out of $5.5\cdot 10^7$ candidates. With learned timing windows and credit to the latest instant, the composition task a Transformer led reaches 0.990-0.999 (mean 0.9965) from 40k examples seen once, vs 0.9955-0.998 for a Transformer after 2M (§88-§89).
- **Grokking occurs, by a route change under sleep, and only for relations the substrate can express.** Without sleep the network memorizes; with sleep it generalizes after a delay (0.93-0.99, reliable with cooled timing noise); a data $\times$ sleep phase diagram shows memorization, grokking and collapse; with depth, $(a + b + c) \bmod p$ through two composed rhythm stages reaches 0.99-1.00 (3 seeds, $p = 17$ and 31).
- **Learning cost follows activity, not model size:** 12 $\rightarrow$ 96 input channels leaves learning mistakes flat and makes inference cheaper (§77).
- **Not yet: real asynchronous benchmarks.** SHD below dense baselines; on the market, correctly posed as trading with costs, no learner profits and the native one learns to stay out; the online world model, built as an event network, beats a neural point process by 0.5-0.9 nats per event, also on held-out days.
- **Transformers are a limit of these networks, including their training, and time adds what they lack (§96, §101-§106).** Races of random clocks compute softmax attention and its gradient on average from local quantities; event networks are controlled differential equations whose universal features are the order detectors they learn; content-dependent delays compute softmax attention exactly at a cost set by its sharpness; a network laid out on positions and time scales is exactly equivariant to shifts and tempo changes, the two ways speakers differ. Tests on real data (spoken digits, language) are queued (E74-E76).

Where the event paradigm wins, and where it does not

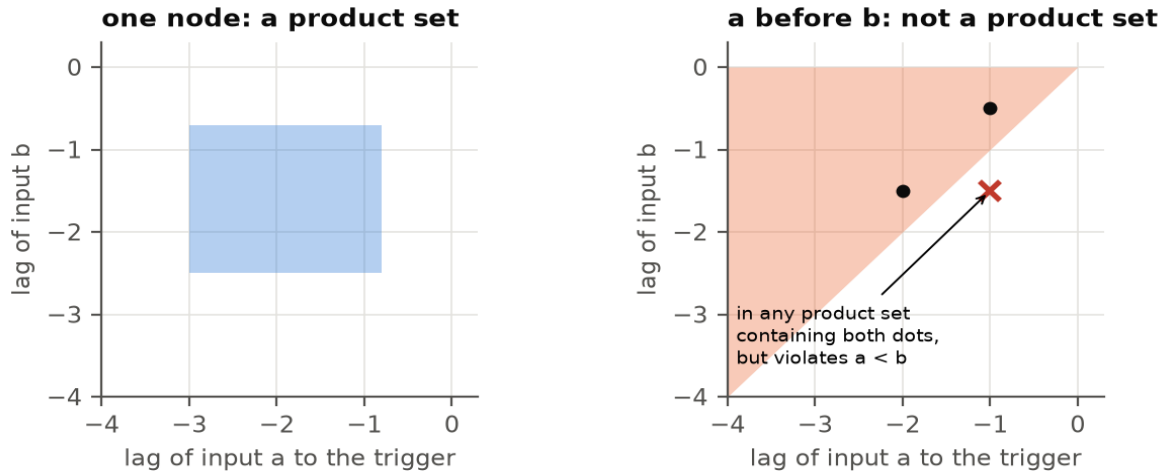
“Supremacy” here means a measured advantage over dense networks given the same data; each row states its caveat.

claim	evidence	caveat
Equal or better accuracy at 10^4-$10^5\times$ lower cost on timing tasks	E35: 1.000 at 7.5 synaptic events, nothing given; conv net 0.995 at 3.07M MACs; event-token Transformer 0.989–0.996 at 146k–1.16M after $10\times$ more training	one task family; much of the gap is the clock (an event-driven conv net $\approx 10\times$)
Groks composed arithmetic where a Transformer does not	E41, $(a + b + c) \bmod 17$, 30% of triples: 0.994–0.999 (3/3) in 200 epochs; Transformer (AdamW, weight decay, 100k steps): 0.29 / 0.63 (seed 0, $d = 32 / 64$), 0.06 / 0.03 (seed 1; chance 0.06)	the two-stage rhythm route is a provided resource (E45: it can choose among routes); more steps might help the Transformer
World model: beats a GRU; within 0.08–0.19 nats of a Transformer point process at $\approx 1/3000$ of its cost	E48: online -2.11 vs GRU -2.62 nats/event; held-out frozen $-2.38 / -2.10$ vs $-3.15 / -2.98$; ≈ 19 synaptic ops vs thousands of MACs	the Transformer Hawkes process is more accurate (held-out $-1.97 / -1.82$ vs $-2.16 / -1.97$, same hazard family)
Learning cost follows activity, not model size	E35: $12 \rightarrow 96$ channels, 0.999–1.000, mistakes flat, inference cheaper ($7.5 \rightarrow 3.2$ – 4.0 synaptic events); \$77, \$81	measured to 96 channels
Structure discovery, implicit Occam razor	E45 pilot: one rhythm for $a + b$ (1.000), the chain for $a + b + c$ (0.999), nothing for random tables	pilot, 2 seeds; 5-seed runs queued
Learned retrieval from $\geq 250\times$ less data	E61: race attention with a learned query-key match, 5/5 runs 100% after 1–4k examples, 100% on $4\times$ longer contexts; of 7 Transformer configurations only $d = 128$, 4 layers solves it, after 400k–1M examples ($4\times$ length: 0.25 absolute positions, 0.72 ALiBi)	the event learner's candidate routes are (item, offset) pairs (ALiBi gives the Transformer relative offsets too); one run per configuration, two for the best (solved at 400k–700k and 700k–1M)
Deep order learned from few examples	E54: 20 orders of four motifs, 0.999–1.000 (5/5) after 10–15k examples, $\approx 2,000$ updates, ≈ 150 events, $80k$ of $5.5 \cdot 10^7$ candidate synapses grown	depth 3: a Transformer matches (0.996–0.999) at $\approx 5,000\times$ the computation; depth 4: 0.990 at equal data, 0.992–0.996 with 2M (4 – $10\times$ the error rate)
Composition: Transformer-level accuracy from one pass	learned windows + latest-instant credit 0.990–0.999 (mean 0.9965) from 40k examples once (E89); Transformer 0.9955–0.998 after 2M, 0.9935–0.9965 given the same $40k \times 50$	one seed at 0.990; dips without a margin
<i>Not supremacy</i> : spoken digits (SHD)	E59 class-conditional event world models, speaker-relative bands, selected on held-out speakers: 0.675 test (E51 0.647); LSTM ≈ 0.70 ; state of the art 95.9–96.3% (event-by-event state-space models)	unseen test speakers
<i>Not supremacy</i> : trading profit	E42 (21 unseen days): no learner beats buy-and-hold (+932 bp); the priced native one +226 bp, others lose	the predictable edge is ≈ 1 bp per trade, below any taker fee (E55, E55b): staying out is correct

1. What an event node computes

Primitives. Spike times; delay; first-of (min) and all-of (max); hold (an input opens a window of given duration); veto (an input blocks the node while within its window). A node fires at its trigger if its holds are met and no veto is active.

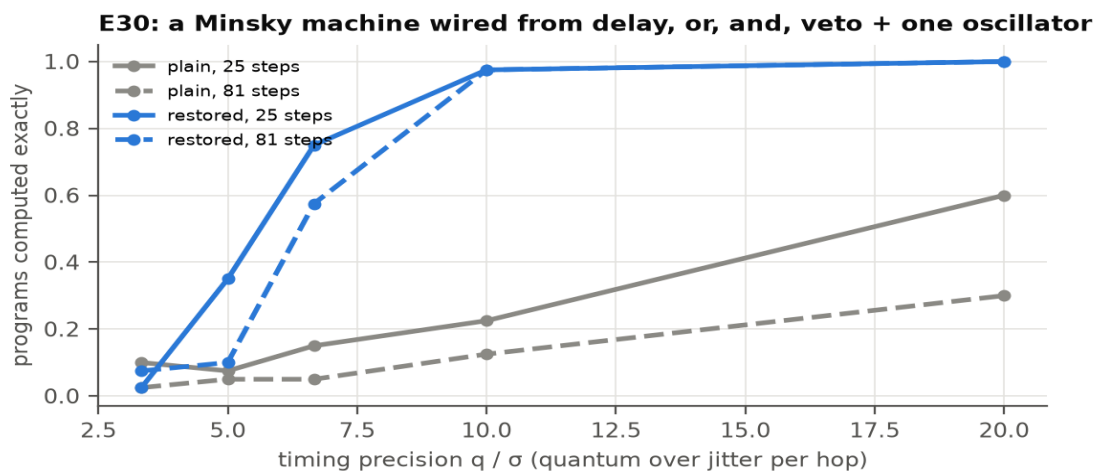
Theorem (§71). In lag coordinates relative to the trigger a stateless node accepts a product set. So one node orders at most three events (trigger the middle one, hold the first, veto the last until it arrives); two layers compute every conjunction of bounded differences with exclusions (every zone of timed-automata verification); three layers every finite union. *Evidence:* an exhaustive search over single nodes finds the proof's own node for 3 events and none for 4 on grids of 1,680 and 11,880 configurations; all thirteen Allen interval relations built at depth 2 are exact on ~19,900 interval pairs (E39).



Stateful nodes are stronger. A node armed by A, disarmed by C and fired by B computes an XNOR of two order relations, which no stateless node computes (exhaustive search, E39b); a toggle node computes parity.

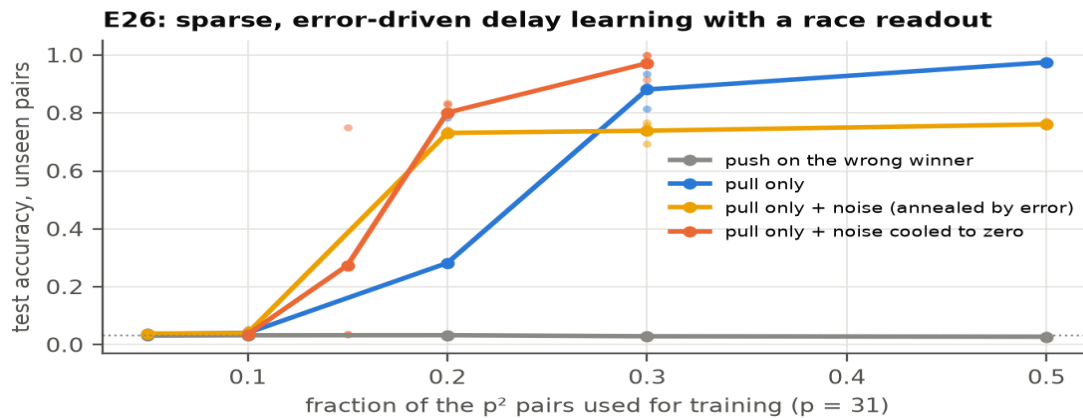
Clockless computation computes relations, not sums: shift symmetry forbids adding two times; one shared rhythm makes cyclic arithmetic computable (§56).

Completeness and reliability (E30). A two-counter machine wired from these primitives plus one oscillator runs exactly (Turing-complete given timing precision). Jitter accumulates like a random walk; one restoring coincidence per cycle makes reliability independent of program length.



2. How event networks learn

Winning is positional (§54). Pull the event that should have won; never push a loser later, which hands the win to the next loser and collapses the detectors. With learned delays on $(a + b) \bmod p$: push at chance in all 15 runs, pull-only 0.97–0.98 (E26). Timing noise cooled to zero makes small-data generalization reliable.



Order needs held intervals, not aligned delays (§61). Delaying A to meet B compresses the interval so a veto cannot see where C fell; A opening a window that B must hit keeps it. With holding, the detector for “B within Δ after A unless C” reaches 1.000 on 5 seeds with ~450 updates in 200k episodes, and veto is worth 19 points (alignment: 0.914, veto worth 1.8).

The rules are online learning on the simplex (§68): pulls moving a fixed fraction of a conserved budget are Winnow/Hedge-type updates with $O(k \log N)$ mistake bounds; durations are interval learning, vetoes monotone disjunctions. With nothing given, detectors learn channels, durations and vetoes: 1.000×4 , 0.9995, 443–1,530 updates (E35).

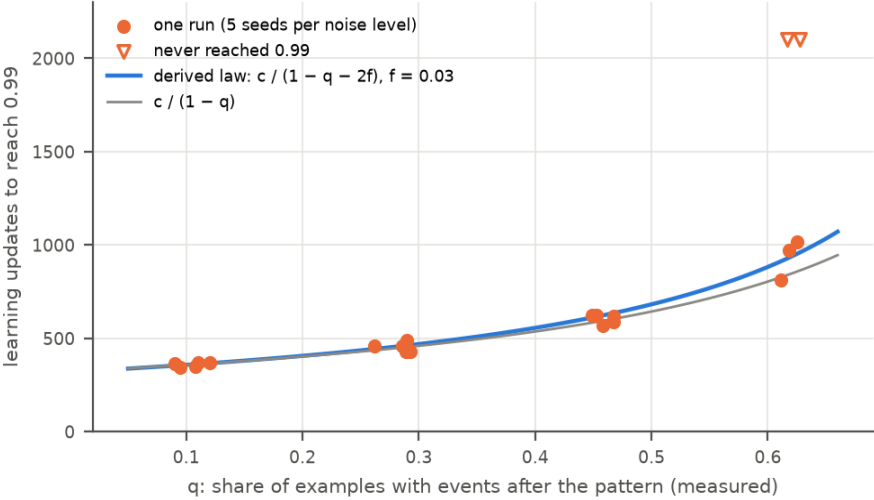
Routing needs counterfactuals (§57). Credit along a spike's causal path cannot say whether another route should have been taken; cancelled near misses supply it. Without it a two-layer network stays at chance (0.17 vs 0.72–0.74, E28).

A learning calculus for deep event networks (§83–§88). Learning deep order natively needed five rules, each derived from a failure, each local to a node and paid for by events:

problem	rule	why it works	evidence
an AND fails: which half was wrong?	sum inputs; multiplicative credit under a conserved budget	conservation moves weight toward the target on every false fire iff $\theta > \frac{1}{2}$ budget (§83, proved)	$4 \times$ candidates $\rightarrow 1.3 \times$ updates (E34w)
candidates fire at different instants	credit one instant, chosen by cooled exploration	all-instant credit deadlocks (proved); greedy cycles on shared prefixes; valid routes absorb (§84)	depth 3: 0.998–0.999 vs 0.71–0.85 greedy, 0.33 all-instant (E53)
the candidate basis grows as P^{depth}	grow a synapse when first credited	exactly the decisions of dense weights (§85, proved, checked)	depth 4: 80k of $5.5 \cdot 10^7$ grown, 0.999–1.000 (E54)
converged nodes sit on their threshold	near-miss margin, earned by recent precision	margins survive r demotions (§86); unearned margins protect wrong routes (§86b)	0.997–0.999 at every checkpoint (E53g)
timing precision	tune one window per part from its own lags	a window bank costs activity quadratic in resolution; tuning costs none (§88)	learned = hand-tuned windows (E56)
first-to-fire commits on partial evidence	credit the latest candidate instant on a miss	the pattern is complete only at its last event; earlier instants are prefixes (§89)	composition 0.990–0.999; depth 3 at half the updates (E89)

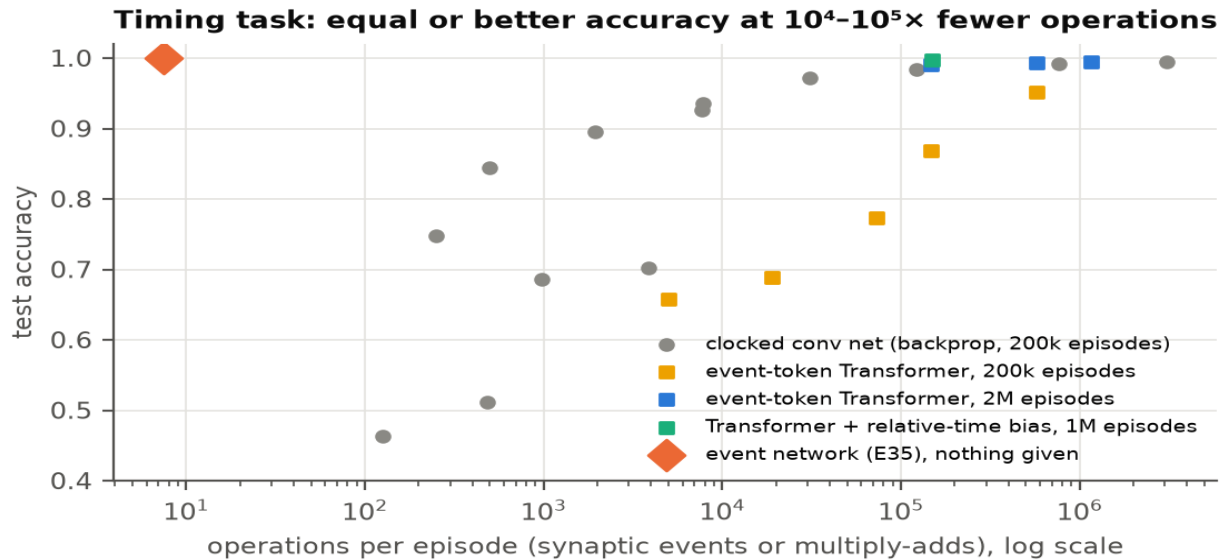
A derived law, measured (§91). Latest-instant credit works because noise after the pattern is inconsistent between examples; the bound says learning slows as $1/(1 - q - 2f)$, q the share of examples whose last event is noise. Sweeping the noise and measuring q , the updates needed follow the law with one fitted parameter ($f \approx 0.03$):

Crediting the latest instant: learning slows as $1/(1 - q - 2f)$, as derived (§91)



3. Against dense models and Transformers

model	test accuracy	cost per episode	learning
event network, nothing given (E35)	1.000 ×4, 0.9995	7.5 synaptic events	443-1,530 updates, 200k episodes
event-token Transformer, 2M episodes	0.989-0.996	146k-1.16M MACs	backprop
Transformer + relative-time bias, 1M episodes	0.996, 0.998	≈ 150k MACs	backprop
event-token Transformer, 200k episodes	0.65-0.97	5k-576k MACs	backprop
clocked conv net (E32)	0.995 / 0.984 / 0.895	3.07M / 123k / 1.9k MACs	backprop, 200k

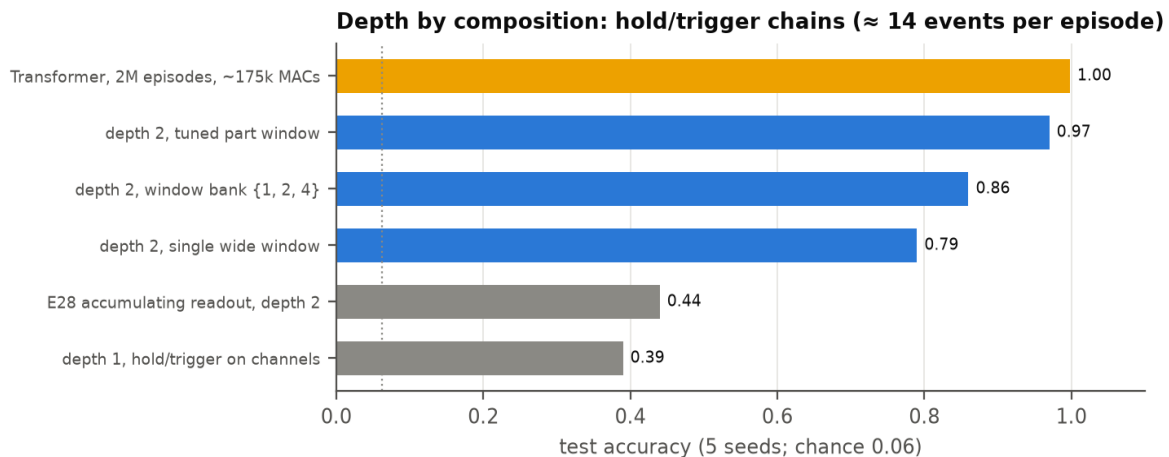


Scaling with the input basis (E35, §77). At fixed spikes per episode, 12 → 96 candidate channels leaves accuracy ≈ 1.0 (96: 1.000, 0.999, 1.000) and learning updates flat while synaptic events per episode fall: learning and inference cost follow activity, not the size of the basis.

Why. Attention cannot see order without position information and must synthesize time comparisons from dot products at $O(n^2 \cdot d)$ per layer; a hold/trigger node computes the comparison as its primitive, at the cost of its input events (§70). The clocked model also pays per time bin, so silence multiplies its cost; a conv net evaluated only where spikes are would cost ≈ 100 multiply-adds, not millions (§63). At published per-operation energies the gap at equal accuracy is ≈ $10^4\times$. *Limits:* one task built around the primitives; the Transformer needed $10\times$ more training for parity; with a learned relative-time attention bias (time differences enter attention directly) it reaches 0.996-0.998 at ≈ 150k MACs, still below the event network at ≈ $2 \cdot 10^4\times$ its cost.

4. Depth and composition

Classes that are ordered combinations of shared parts are zones over part events, so the theorem prescribes two layers: parts (hold nodes on channel pairs) and class nodes that hold one part's spike and are triggered by another's, so a single part cannot satisfy them. Learned natively: 0.97 with part windows at the motif scale, 0.86 with a generic bank of window scales, depth 1 0.39 (5 seeds, E34). A readout that accumulates evidence fails at conjunction even though its hidden nodes learn parts (E28). An event-token Transformer trained on 2M episodes reaches 0.998 at $\approx 175\text{k}$ – 690k multiply-adds vs ≈ 14 events. The chains are not training-limited: at 200k episodes they stay at 0.954 (tuned) and 0.87 (bank); some class routes lock onto wrong parts.



What the credit rule must be (§83–§84). A class node should sum its held and its coincident trigger inputs and learn by full-information multiplicative updates under a conserved budget. Proved: conservation resolves the AND's credit ambiguity (a false fire does not say which half was wrong) iff the threshold exceeds half the budget; the mistake bound grows with the log of the candidate basis ($16 \rightarrow 32$ channels: $1.3\times$ the updates). Crediting every candidate deadlocks when units fire in every positive at different instants (window bank: 0.05–0.10); crediting the one instant closest to firing can cycle between invalid prefixes; cooled exploration over instants reaches the valid route, which is absorbing. With a class window covering the task's span, E34's task plateaus at 0.988–0.995 per seed and the generic bank at 0.97–0.99 with fixed windows. **Learned windows (§88) plus credit to the latest instant (§89)** remove the last errors, lost races to a shortcut that fires when the second motif merely begins: 0.990–0.999 (mean 0.9965) from 40k examples seen once, against a Transformer's 0.9955–0.998 after 2M examples and 0.9935–0.9965 given the same 40k examples 50 times: parity at equal data, from one pass, at $\approx 10^4\times$ less computation.

Order among three parts: depth 3 (E53, E54). Classes that are different orders of the same motif sets (20 classes, decoys = other orders) need ordered intermediates: composite units $u \rightarrow v$ over every ordered pair of parts (57,840 candidates, ≈ 41 events per episode). 5 seeds, 40k episodes:

E53	test accuracy	updates
depth 3, + margin earned by reliability (§86b)	0.997–0.999 at every checkpoint, 5/5 seeds	$\approx 1,100$ – $1,400$
depth 3, instant credit, exploration $T = 0.3$	0.998–0.999 on all seeds by 5k episodes; final 0.963–0.999 (dips)	$\approx 1,100$
depth 3, greedy instant credit	0.71–0.85 (cycles between shared prefixes)	4k–11k
depth 3, credit to every candidate	0.33–0.36 (never fires)	$\approx 26\text{k}$
depth 2, same credit	0.32–0.42	$\approx 24\text{k}$
Transformer, 2M fresh examples	0.997–0.999	215k–843k MACs/example
Transformer, 40k $\times$ 50 passes (wd 0 / 0.1)	0.9975, 0.996 / 0.981, 0.994	215k MACs/example

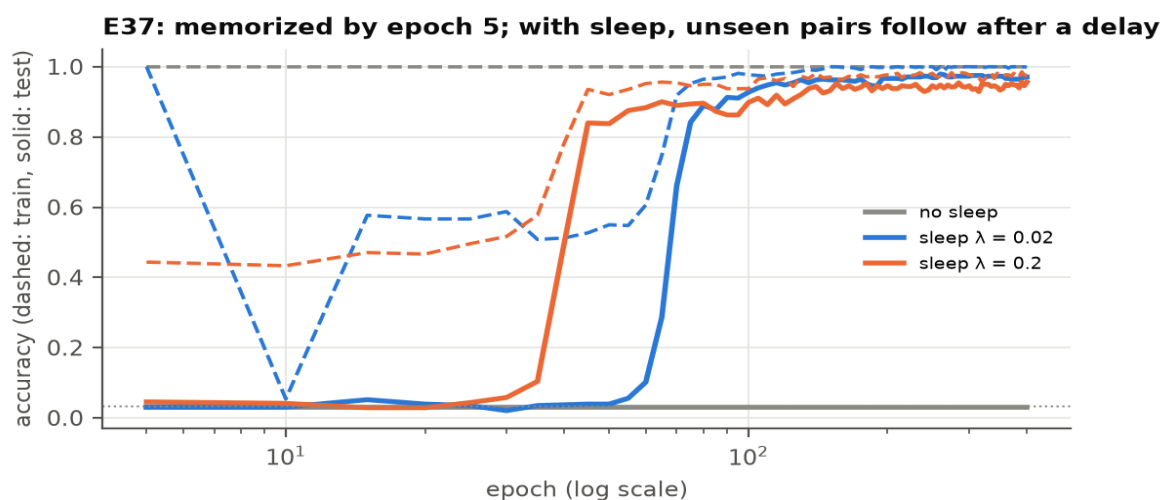
Without a margin, converged classes dip (a node sits just above threshold; one demotion knocks it under). A margin kept by near-miss credit removes the dips but also protects wrong routes (3/5 seeds

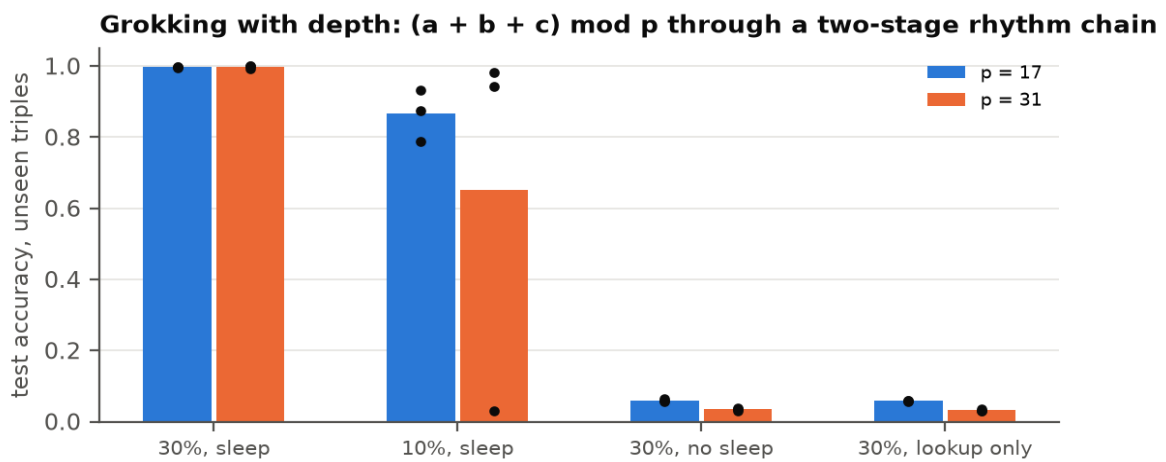
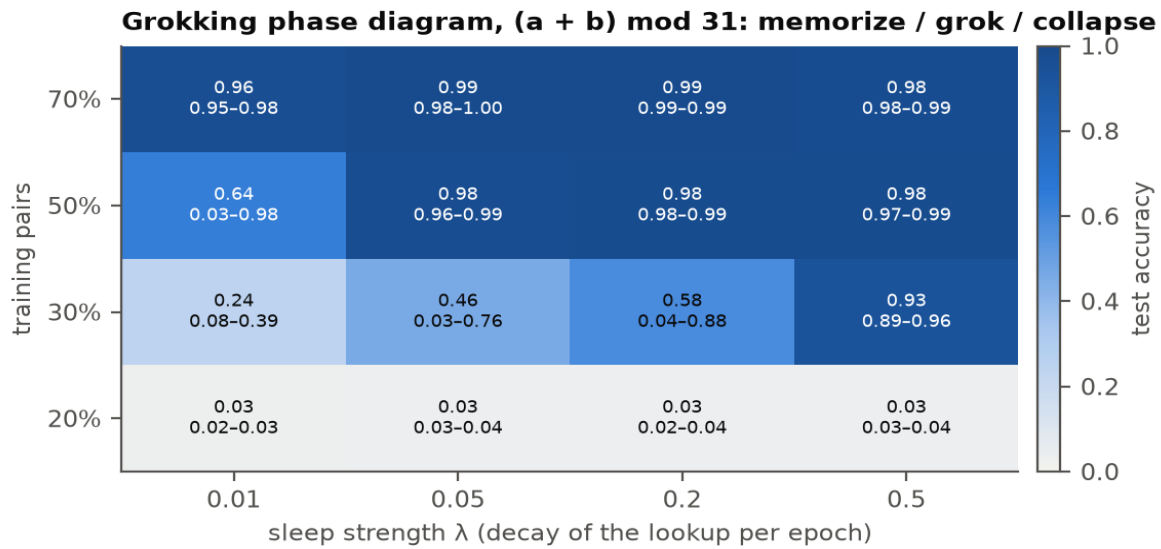
freeze on a shared prefix); a margin earned by reliability, applied only when the node's recent fires were mostly correct, is stable at 0.997–0.999 (§86b). With synapses grown only when first credited (§85, provably the same decisions as dense weights), depth 3 at 20 channels gives 0.981–1.000 with 9.5k–13.7k of 145k candidate synapses ever grown. **Inference cost follows the learned structure (§93):** extending a unit only if one of its children carries weight (checked periodically, like sleep) keeps accuracy (depth 3 identical; depth 4 0.995–1.000) while events per example fall 42% at depth 3 and 75% at depth 4 ($\approx 155 \rightarrow 39$).

5. Generalization and grokking

What counts (§58): restriction, forced generalization above capacity, and grokking (the relation reached while memorizers are available) are different claims; each reports $\rho = n/\text{params}$. **Per-class parameters cannot generalize on $(a + b) \bmod p$ (§66):** each operand occurs once per class, so generalization needs shared intermediates; in time, a sum needs a rhythm.

Grokking as a route change (E37). Each class has a pair-node lookup that can memorize everything ($\rho \approx 0.016$) and a shared route through a rhythm with learned delays; learning is errors-only. Without sleep: train 1.0, test 0.03–0.04. With sleep: test 0.93–0.97 in 2 of 3 seeds without timing noise, and 0.95–0.99 in 5 of 5 seeds with cooled timing noise on the shared route ($\sigma = 2$), after a delay. Sleep without the rhythm: train 0.45, test 0. The rhythm is used only where it fits: $a - b$ and relabelled sums grok (0.95–0.98), $a \cdot b$ in 1 of 3 seeds, while $a^2 + ab + b^2$ and random tables stay at chance on unseen pairs (0.01–0.04) and their training accuracy erodes under sleep. A data $\times$ sleep phase diagram (4×4 , 3 seeds) shows no grokking below 20–30% of pairs, and above it a minimum sleep that falls with data. Larger problems grok more reliably: 2/3 seeds at $p = 31$ and 59, 3/3 at $p = 97$ ($\rho \approx 0.005$). With depth (E41, 3 seeds): $(a + b + c) \bmod p$ through two composed rhythm stages, 0.99–1.00 on unseen triples from 30% of them at $p = 17$ and 31, and from 10% in 2 of 3 seeds at $p = 31$ ($\rho \approx 0.003$); chance without sleep; collapse without the chain. The data it needs is far above the Occam bound (§78): at 1–7% of triples it memorizes and stays at chance, also when trained 7–10 $\times$ longer; the threshold at $p = 17$ lies between 7% and 30%.



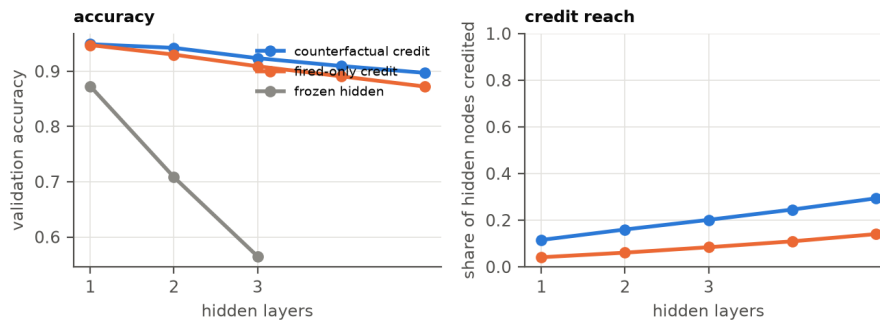


Why (§69, §72). Error-gated learning makes memorization absorbing. Sleep keeps a parameter only if it is used by more than $m^* = \lambda\theta/(e\eta)$ examples: lookup entries serve one and die, the rhythm's delays serve many and survive, and once the rhythm answers a pair its lookup entry is never relearned. This predicts memorization, grokking and collapse regimes (all observed) and a data threshold $n^* \propto p\lambda\theta/(e\eta)$.

6. The weight race

The original architecture (integrate-to-threshold nodes with learned weights, racing in groups) holds its first results: a cancelled node can be taught (E4: 0.79 at $K = 128$ where reward-modulated rules are at chance); races decide as fast as the evidence allows (E2, matching the optimal MSPRT); learning work tracks activity (E5: $\approx 300\times$ fewer updates); local learning reaches ≈ 0.96 on latency-coded MNIST with counterfactual credit that pays more with depth (E6, E14).

E14 · the advantage of counterfactual credit grows with depth (2 seeds)



Its limits: depth still costs accuracy; only its single racing layer beats an equal dense model at inference ($\approx 2.4\times$); it forgets more than SGD (E23); it memorizes instead of grokking (E24); and the rules that fixed the timing networks do not transfer to its weights (SHD 0.04-0.29 vs 0.35).

7. Real data

- **Spiking Heidelberg Digits** (spoken digits as cochlear spike trains, 700 channels, 20 classes, unseen test speakers): class-conditional event world models (E51; state = last spike's band, time since it, time since onset; one counting pass) reach 0.647 test (0.734 on held-in speakers); timing +0.06, onset reference +0.21. The weight race reached 0.35; a published LSTM ≈ 0.70 ; the state of the art is 95.1% (learned delays), 95.9% (Event-SSM) and 96.3% (S7): the last two process spikes one event at a time with linear state-space units, which §104 shows are event units of our kind with every unit updated on every event (both select checkpoints on the test set); time only fades their state, so they do not compute with delays. E74 (queued) tests the paradigm's own design: events carry small vectors whose content sets their delays and whether they are sent; the receiver's clock weights and rotates what arrives; units fire at threshold and emit their state at that moment (§105). SHD is only $\approx 6\times$ sparser than a 10 ms raster, a weak test of the paradigm's cost advantage. Validating on held-out speakers and coding bands relative to each voice (a running centroid per utterance) raises held-out-speaker accuracy from 0.36-0.38 to 0.44-0.46 and the test to 0.675 (E59, §92).
- **Market stream posed as trading with costs (E42, confirmed on 21 unseen days, preregistered)**: at 2 bp, imitating a hindsight teacher over-trades and loses (event learner -15,236 bp, logistic -24,302); the profit-priced event learner nets +226 bp with 8 changes; buy-and-hold +932; at 10 bp all stay out. No learner beats buy-and-hold; pricing the decision is what stops the losses.
- **Is staying out right? An edge audit (E55, §87)**. From executable round trips on the tape (buy at the ask, sell at the bid), BTC spot's own event states carry a real held-out edge of +0.3 to +0.9 bp per trade before fees, and lead-lag states of BTC perpetual futures and ETH raise it to +1.1 to +1.5 bp: more markets carry more information. A 2 bp round-trip fee (a tenth of a realistic taker fee) removes it: staying out is correct for a taker here; the edge would need market-making economics. **Confirmed on the 21 untouched days** (preregistered): before fees +0.26 to +1.05 bp from the own state, +0.47 to +1.40 bp with perp and ETH; at 2 bp every selected state loses; at 5 bp none qualifies. **Across four markets** (ETH spot, SOL spot or the BTC perpetual traded, the others as leaders; choices made on the pilot days): at most ≈ 1 bp before fees, nothing at 2 bp (E55b).
- **Event-camera gestures (DVS128 Gesture, E60; official split)**. Native motion events (refractory cells, onsets, direction-selective pair detectors with an opponent veto) turn $\approx 410k$ raw events per gesture into $\approx 15k$. A bag of motion events reaches 0.663 on unseen people; depth (successive motions per region) with the multiplicative learner reaches 0.776 on held-out training users and

0.701 on the test users (chosen on validation). Published systems reach 94–98%: not yet competitive; the representation misses rotation sense and trajectory shape.

- **Against a Transformer point process (E52, E57, §90).** A Transformer Hawkes process with the event network's own hazard family and 128 events of context, selected on day 5, scores $-1.97 / -1.82$ nats per event on the held-out days ($\approx 108k$ multiply-adds per event; $-1.83 / -1.65$ with 12 finer windows). The semi-Markov event network scores $-2.38 / -2.10$; slow regime state (leaky event counters at 5 s and 60 s, an order-flow counter, backoff) brings it to $-2.18 / -2.00$, and per-type counters as a third backoff level to $-2.16 / -1.97$ (fine windows $-1.91 / -1.73$; chosen on day 5) at ≈ 30 operations per event. The Transformer is the better world model by 0.08–0.19 nats; the event network gets within that at $\approx 1/3000$ of the computation. Counted slow state transfers to unseen days; constant-step multiplicative factors track the end of training and do not (E58).
- **The world model is an event network, and it beats a neural point process (E44, E48).** A likelihood decomposition located the GRU's lead in which event comes next; count baselines located the missing information (the time since the last event). A semi-Markov event network (state nodes for the last two types, window nodes from a delay line, count-learned detectors) reproduces that model exactly and scores -2.11 nats per event online (days 1–5) and $-2.38 / -2.10$ frozen on held-out days 6 / 7, against -2.62 and $-3.15 / -2.98$ for the GRU, at ≈ 19 synaptic operations per event. The model class is classical; an offline-trained GRU scores $-2.72 / -2.53$ on the held-out days, also behind (E49); the Transformer point process is ahead of both (above).
- **Earlier native world model (E44), prequential log-likelihood per event (nats; days 1 / 2 / 3):** Poisson $-3.00 / -3.42 / -3.32$; Hawkes (Adam) $-2.62 / -2.94 / -2.84$; native (multiplicative) $-2.64 / -2.85 / -2.70$; GRU neural point process $- / -2.61 / -2.52$. Pair-part state neutral; learned inhibition below excitation-only; the semi-Markov network of E48 (above) closed the gap to the GRU.

8. Open problems and next steps

- **Stability of the full rule set on every task at once:** the margin earned by reliability is stable at depth 3–4 and with fixed windows but hurts when windows are learned (it entrenches early shortcuts at the firing instant and promotes trailing noise at the latest instant, §89): the margin needs another anchor.
- **Depth beyond four and denser streams:** depth costs activity $n \cdot r^L$ (§85); extending a unit only toward children that carry weight cuts events by 42% at depth 3 and 75% at depth 4 at unchanged accuracy (§93); depth 5 is queued.
- **Time-vector networks on real streams (§105–§106):** E74 (content-dependent delays, snapshot payloads) and E75 (the same, exactly equivariant to band shifts and tempo) on spoken digits, selected on held-out speakers: do delays and vectors computing together close the gap to 95–96%?
- **The work law of attention in language (§106a):** how many keys a trained character-level Transformer's queries actually need as the context grows (E76), which fixes what delay-coded attention saves on text.
- **A time-vector language model (§107, E77):** does sparse time-vector memory plus delay-coded retrieval reach the converged LSTM and Transformer at equal data, and how many keys does it retrieve per character?
- **Structure discovery for grokking** (E45 pilot picks correctly) and **the data threshold of grokking** (7–30% of triples, far above the Occam bound; the sleep reuse filter is the candidate constraint).
- **Native learning of sparse parity (§75) and a real benchmark with rare, precisely timed events (§55).**
- **Joules, not operation counts:** run trained networks on neuromorphic hardware (§9).

9. Hardware: what these networks need, and what exists

Each primitive the theory settled on maps to a hardware feature:

what the network does	what hardware must provide	why
work only when an event arrives	event-driven execution; memory next to compute	cost follows activity (\$55, \$77)
delays and hold windows	per-synapse programmable delays, per-node hold timers, timestamps	order is held intervals (\$61, \$71)
first to fire wins, the rest cancelled	fast arrival-order resolution and inhibition	the race readout (\$59)
multiplicative credit under a conserved budget	per-synapse multiply, per-node renormalize, contribution tags, local random source, precision counter	\$83–\$86b, \$89
synapses grown when first credited	run-time allocation in a sparse synapse store	basis 10^5 – 10^{10} , 10^4 – 10^5 grown (\$85)
part windows tuned from their own lags	per-synapse window edges with a local rule	\$88
signals carrying small vectors whose content sets their delay	8–32 values per event, a small dense core per unit, a delay computed per message, jitter small against the delay scale	delay-coded attention (\$105); latency $\times$ weight error = jitter $\times$ logit range (\$106b)

The optimal machine (a sketch). Clockless digital cores with timestamped events, per-synapse delay and window fields and per-node timers; arrival-order comparators and inhibition trees for the race; a small event-triggered learning engine per core; a content-addressed sparse synapse store with allocation on credit. This is close to *race logic* (first arrival = minimum, delay = addition: the max-plus algebra of §79) plus learning. At ≈ 24 pJ per synaptic event (Loihi, 2018) these networks would spend ≈ 0.2 nJ per example on the timing task, ≈ 0.5 nJ on composition and ≈ 4 nJ at depth 4; a Transformer at 175k multiply-adds per example on a GPU is on the order of a microjoule (orders of magnitude, not measurements).

What exists (September 2026):

system	availability	fit
Intel Loihi 2 / Hala Point	research access; Hala Point a prototype (Sandia); Loihi 3 announced, no public specs	best for prototyping: event-driven; weight, delay (≤ 62 steps), tag per synapse; microcode learning rules. Missing: run-time synapse allocation, long delays, timestamps
SpiNNaker2 (SpiNNcloud)	commercial systems; 152 ARM cores/chip	most flexible: every rule incl. synapse growth in software; less efficient per event; time-stepped
BrainChip Akida / Akida Pico	commercial; Pico in FPGA-cloud evaluation (2026)	converted CNN-style SNNs, limited learning, no suitable delays: poor fit
Innatera Pulsar	volume production (2026), μ W–mW	deploy small trained networks at the sensor
SynSense Speck / Xylo	commercial dev kits	inference-only spiking ASICs for vision / audio
DYNAP-SE2, BrainScaleS-2	research	analog continuous time; device mismatch
IBM NorthPole	research	synchronous dense inference: not a fit
FPGAs	commercial	delays as timestamp queues, synapses in BRAM hash tables: the most faithful full implementation today
event sensors (Sony/Prophesee, iniVation)	commercial	natural front ends: they emit these event streams

What to do with it. Measure joules: trained networks on Loihi 2, the full learning calculus on SpiNNaker2 or an FPGA, against a Transformer on a GPU; deploy frozen networks on sensor-edge chips. The feature no commercial chip offers, and the one these results say matters most, is **run-time synapse allocation on credit** with per-node conserved budgets: it lets a network search 10^7 – 10^{10} candidates while storing only what it uses. Sources are listed in REPORT.md §9.

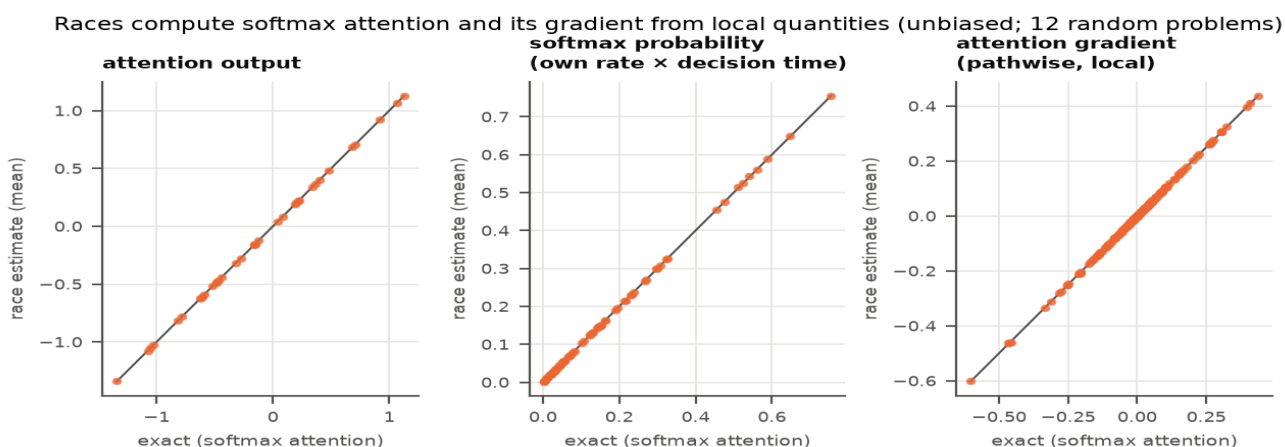
10. Next frontier: generative language models

The aim is not to approximate Transformers but to exceed them: the same or better quality, with work per word that does not grow with model size or text length, learned by local rules from less data.

Why that is a reasonable aim (theory, \$96–\$98):

- **Nothing a Transformer computes is out of reach.** Query-key similarity is the overlap of spike codes; a race among stored keys picks the best match, and a race of randomly ticking clocks picks each key with exactly its softmax-attention probability; relative position is native; the feed-forward block is threshold units over codes; stacking layers is composition. A Sleeping Machines network can express any Transformer.
- **It has freedoms a Transformer lacks:** the order in which signals fire carries up to $\log_2 n!$ extra bits for n signals; only active units work, so a model can be very large while each word stays cheap; structure grows where it proves useful; sampling is a race; retrieval reaches only keys sharing a channel with the query.
- **Local learning is not a handicap in principle.** A race computes with minima and sums; the exact gradient backpropagation would compute runs only along the chain of spikes that caused the output, which each node traces locally: credit along that chain is backpropagation for these networks, and near misses supply the signal gradients cannot give to losing paths. For races of random clocks the exact gradient is local too.
- **Depth is trainable, optimally.** For ordered-pattern detectors of depth d , mistakes grow as $d \times \log(\text{candidate pool})$, and no learner can do better in the worst case (§97–§98).

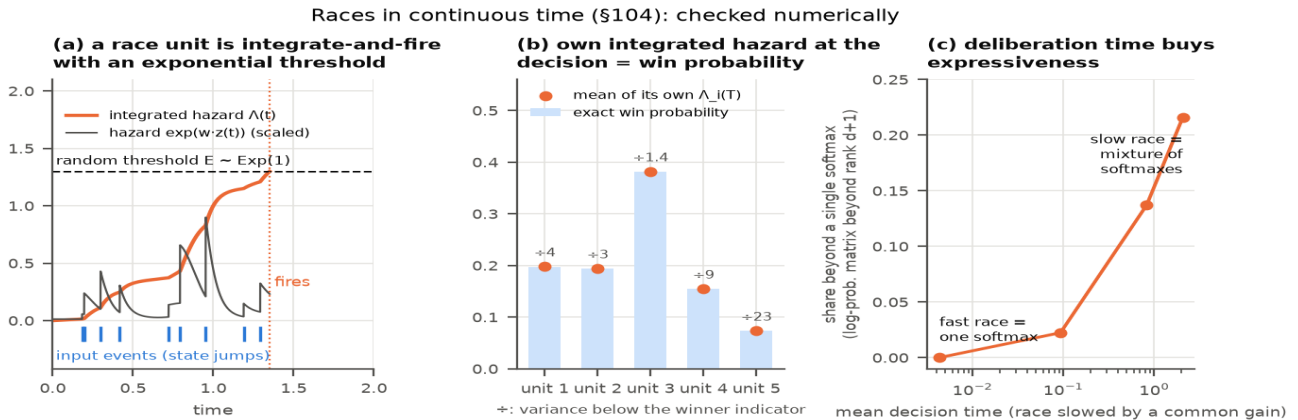
Training, subsumed (theory, §101–§103). A race of randomly ticking clocks splits its output into two independent channels: which clock fires first (a sample from the softmax) and when (the decision time, which carries the softmax's normalizer). Proved: every competitor computes its own softmax probability locally (its rate times the decision time); keys emitting values scaled by that product give, on average, exactly softmax attention; and the locally computed gradient of this race is, on average, exactly the gradient of softmax attention. A network of such races with small dense cores is trained by local message passing as stochastic gradient descent on the Transformer objective, up to an error shrinking as $1/R$ with R races per head: Transformers, including their training, are a limit of these networks. Numerical check below; training curves compared in E68 (queued).



Time and content, one system (theory, §104). Continuous-time neural models describe a hidden state that flows and is pushed by its input: neural ODEs, controlled differential equations, and the state-space models behind Mamba-class language models. An event network is exactly such a system. Between events its state flows in closed form, and at each event it jumps. The only nonlinearity is in *which* units fire *when*. Four consequences follow:

- **Order detectors are the natural features of event streams.** The iterated integrals that make these models universal (the “signature” of a path) are, for event streams, the counts of ordered event patterns: exactly what our “this, then that” detectors learn. Stacking such detectors builds the universal feature set.
- **Selection comes free.** Mamba-class models gain their power by letting the input set how fast the state forgets. In an event network, which channel fired is that signal. A unit that an event does not address need not be touched at all, and skipping it is exact (sleeping execution). The best published models on spoken digits (95.9–96.3%) are such units with every unit updated on every event, and time only fades their state: they do not compute with delays.

- **A race unit is an integrate-and-fire neuron with a random threshold.** Its gradient is the event-based backpropagation used for spiking networks, but the random threshold keeps the expected loss smooth even when spikes appear or vanish. At the moment of decision, each unit's own integral is on average exactly its probability of winning, for any time-varying rates.
- **Time adds expressiveness.** If the scores change while the race runs, the race outputs a mixture of softmaxes over its own decision time. That is more expressive than the single softmax at the end of every Transformer (the “softmax bottleneck”). A fast race is one softmax; a slower race buys expressiveness with time rather than with parameters.



Delays and vectors, computing together (theory, §105). An event carries a small vector, and its content decides when it arrives: a message whose content matches the receiver is delayed in proportion to the match, and one that does not match is never sent. The receiver's state fades with time, so a later arrival counts more. Proved and checked: the receiver holds exactly softmax attention over the matching messages, with no multiplications for the weights and no sampling, paying only for messages sent. Races compute the same softmax by sampling (fast, slightly noisy); delays compute it by waiting (exact, slower for a wider range of scores). A unit fires when its evidence crosses threshold and sends on its state at that moment, so what it says and when it says it are one computation. Such networks compute in the log semiring: delays add, gains multiply. E74 tests it on spoken digits.

Two memories (theory, §107). Such a unit is, algebraically, the exponentially gated memory of xLSTM (a family already shown to scale to billions of parameters competitively with Transformers): elapsed time is its forget gate, the content-dependent delay its input gate, the count channel its normalizer. But a memory that chooses what to keep when it writes cannot answer arbitrary questions asked later: remembering N facts for any future question needs at least $N \times (\text{bits per fact})$ of state. A language model built this way needs a second memory, retrieval, done natively: a question is sent to stored keys, which reply sooner the better they match; the first reply opens a short window, and replies inside it are weighted exponentially, which is exactly softmax attention over the good matches. Aggregating values costs the number of good matches; finding them still costs linear work without an index. E77 is a configurable, deeper-capable prototype; its new adaptive path is not yet measured against the converged language baselines.

Potential: a layer can choose its memory operation. Softmax key/value attention is a one-step modern Hopfield retrieval rule under the associative-memory interpretation. E77 now applies a separate causal query/key/value update to emitted event payloads, then passes the retrieved vector onward through the event stream. Its score gradient is centered on the retrieved value, teaching both where to read and what to send. The temporal layers remain event-state layers; the design is heterogeneous by intent. A learned gate mixes this event-level lookup with an identity payload path. In the auto-associative case, retrieval is the gradient of a convex log-partition function with a positive-semidefinite covariance Hessian; repeated retrieval is contractive when inverse temperature times key-diameter squared / 4 is below one. Separate keys and values enable hetero-association but need not preserve this energy structure, so E77 uses one gated update per layer.

A depth result with explicit assumptions. For fixed keys and values, the query sensitivity of softmax retrieval is a cross-covariance bounded by inverse temperature times key diameter times value diameter / 4, independent of the number of memories. If each event-level correction has bounded full

Jacobian, residual scaling by $1/\text{depth}$ keeps the stack Jacobian bounded away from zero and infinity. E77 logs score spread, retrieval entropy, update size, diameter upper bounds, the query-Jacobian bound, and role-specific gradients. Hard event births and learned route discovery are not covered by the bound.

Precision and learning credit trade off. With score margin m , non-winner mass is at most $(N-1)e^{(-\beta m)}$, so sharper scores improve retrieval. For two keys, however, score sensitivity is $\beta p(1-p)$: it peaks at a tie and vanishes when the route is certain. An excluded key gets no gradient. This mathematically motivates broad early retrieval, near-miss credit, and gradual sparsification.

If these mechanisms scale. Deep event stacks that preserve associative recall and learn useful sparse routes could grow model memory and reasoning capacity without making every token pay for every possible interaction. Training would follow predictive routes; inference would follow emitted events and retrieved candidates. Capability growth would depend less on dense matrix size and more on temporal composition, associative memory, and sparse routing. A stronger accuracy-per-compute curve could shift frontier investment toward fast memory, event-capable processors, and distributed associative stores, while reducing dependence on ever-larger dense GPU clusters.

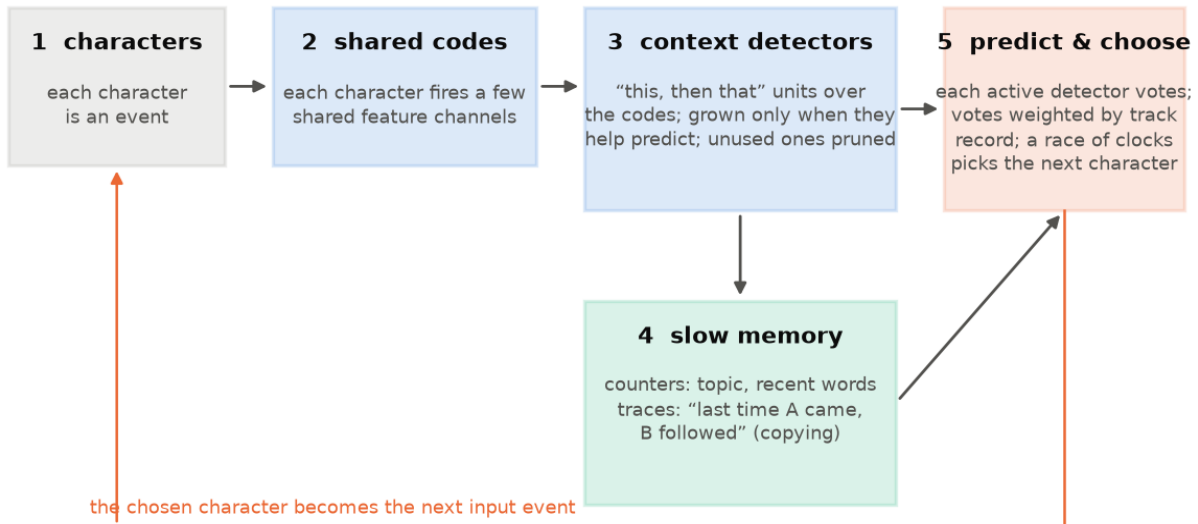
Cost is still a live question. The event Hopfield update runs only at emitted events, but it scores every eligible event pair. Token retrieval also scores every query/key pair, and TVLayer still allocates dense time-by-batch-by-unit state. Sparse wiring is not yet sparse execution or a measured energy win.

Scaling depth. E77 accepts configurable event depth. Each deeper layer receives prior events plus sparse raw-input skips, with event-level Hopfield updates scaled by $1/\text{depth}$; this is not a dense tokenwise residual block. Depth sweeps will track route coverage and optimization; the two-layer configuration is a starting point, not a depth limit.

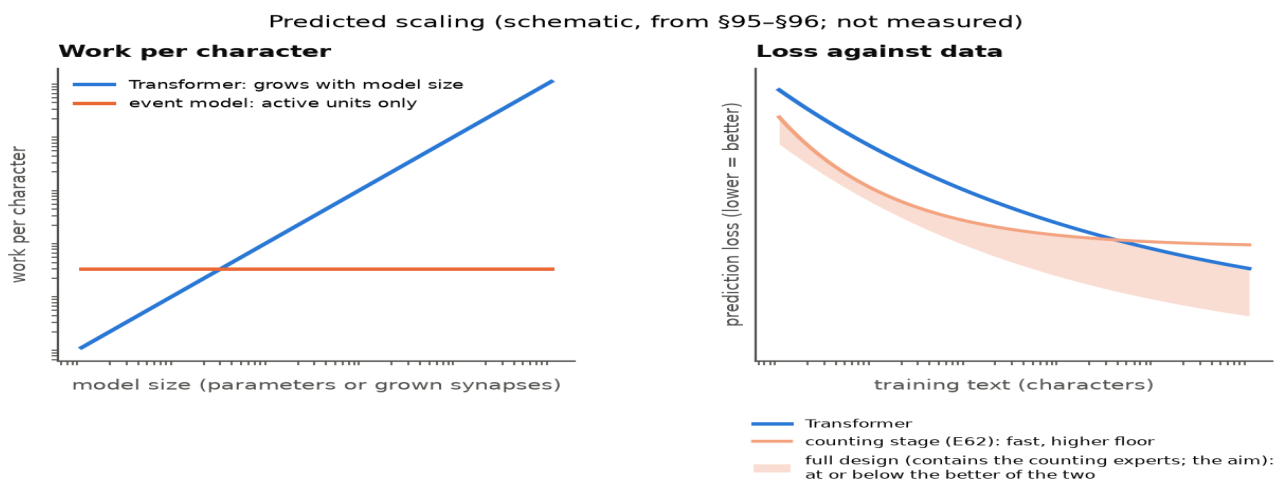
Local learning that provably suffices (theory, §108-§109). When units predict the next character by racing (each candidate's clock rate a weighted sum of the log-probabilities its inputs assign), a network of such units is a gated linear network: every unit predicts the target itself and learns only its own convex loss, so no error is sent backwards, and such networks are known to be universal and to learn well in a single pass (Veness et al., 2021). The network is never worse than its best part. These units do not build features; here features come from the time-vector layers and native detectors, and the race units combine them.

First evidence. Deep order is learned from about ten times less data than a Transformer needs (section 4). Attention is learnable by local credit, from far less data: in a recall task where the network must learn which key a query refers to and which neighbour to read, a race-attention layer trained by local credit alone is 100% correct after 1-4k examples and 64-68 mistakes (5/5 runs) and 100% on contexts four times longer; the Transformers that solve it need 400k-1M examples and reach at most 72% on the longer contexts (E61). **Language, stage 1** (counting experts, copy memories and word-keyed memories, mixed by conserved multiplicative credit), measured on text8 test text at 1M, 10M and 90M training characters: the mixture reaches 2.00, 1.79 and 1.65 bits per character (E63), and word-keyed experts bring it to 1.98 and 1.73 at 1M and 10M (E66). Stored contexts grow as $D^{0.41}$ and pairs as $D^{0.49}$. Counting alone (E62, 2.31 $\rightarrow$ 1.81) fits a floor near 1.73 bpc, which the mixture already passes: the floor belongs to the component, not to the design. **Mixing by a race** (§108: each candidate's clock rate is the weighted sum of the experts' log-probabilities, a product of experts, as the best compressors mix) takes the same experts at 10M characters from 1.80 to **1.61 bits per character**, with weights frozen after the validation text and the copy memory limited to 256 characters (the Transformer baseline's context), at a few hundred operations per character (E79), and at 90M to 1.50 (published text8: standard LSTM $\approx$ 1.43, stronger recurrent models 1.27-1.36, large Transformers $\approx$ 1.08: at full scale near an LSTM, behind Transformers). For scale, large Transformers reach $\approx$ 1.1 on text8 from 90M characters. Our first gradient-trained baselines at equal data (one pass over 10M characters: LSTM 2.17, Transformer 2.43) are not converged, so no comparison is claimed until converged runs (multiple passes, early stopping on validation, E64b) finish.

A planned event language model: work only where characters arrive (design, not yet built)



The plan, in stages, on character-level text (text8): (1) a counting baseline with a copy memory (measured, E62–E66, above), not the goal but a measurement of how memory and loss scale with data; (2) attention over the stream, by races (E61 at scale; E68) or by content-dependent delays (§105: exact, and as cheap as the attention is sharp; E76 measures how sharp a trained model's attention on text is; E77 is the first full model of this kind); (3) learned shared codes; (4) stacked layers with credit along causal chains and near misses. At each stage: a recurrent network and a Transformer trained by gradients on the same text; bits per character, examples needed, work per character.



Established and not. Established: expressive equivalence, locality of exact credit for races, optimality of the depth bound, exact delay-coded attention and its work law (theory, checked), learned attention on a recall task from $\geq 250\times$ less data than a Transformer, data efficiency on deep order. Not yet shown: that stacked race-attention layers with learned codes, trained by local credit, match or beat a Transformer on language itself; the stages decide it.

Every mechanism is an event handler (local state, triggered by events, cost proportional to events); dense procedures are diagnostics only; time-vector networks (E73–E75) are trained by gradients that flow only through spikes that occurred, simulated on a 1 ms grid for speed (an event-driven adjoint form exists). Reproduce: `python report/figures_time.py` && `python report/make_pdf.py`.